

KITEWRIGHT CORE · CALCULATION

Kitewright Core sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First issue for TRL 3 on the constructable design (KWC-DDR-002): mount strength, payload swap, payload power, heat, altitude, hover-time model, common core, mass and cost

Kitewright Core sizing calculations

On paper the constructable core meets ten of its eleven requirements and misses one. The plain rail and two locking pins carry the 147 N vertical design load with factors of 10 or more on every part, a gloved payload swap takes about 52 s with no tools, 100 W reaches the payload at no more than 5.6 A, and the hottest avionics part stays near 75 °C at 45 °C ambient. The hover-time model explains the scaffold's 47 % figure: at 5,000 m and -20 °C an aircraft keeps 77.5 % of its sea-level hover time from air density alone, and unheated packs that give 60 % of their capacity bring it to 46.5 %; ColdCell packs kept warm raise it to 70.0 %. The core's estimated mass, 1.27 kg, does not meet R9 (Decisions for Amish in `docs/REVIEW.md`). The estimated cost of USD 3,940.60, with ground station and spares, is USD 1,059.40 under the USD 5,000 value-engineering target. Every number here is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that the payload mount, the lithium packs, the power wiring or the aircraft is safe. The mount must be proof-loaded and the locking pins pull-tested, and the power board must be checked on a current-limited bench supply, before any pack is connected or any payload is flown. See KWC-PRC-001, Safety.

■ Scope and method

The note checks every requirement in KWC-REQ-001 v0.2 against the design in KWC-PRC-001 v0.2 and the parametric model `cad/src/model.py`. The script imports the model's parameters and part volumes, so the plate, rails, shoe, pin blocks and lid used here are the ones in the STEP file, drawing KWC-DWG-001 and the build plan. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Rated payload	5 kg on the mount, one payload class for version 1 (KWC-DDR-001, D4)	Kitewright Lift's sea-level payload
Load cases	Static 3 x 5 kg; drop 1.5 x 5 kg at 2 g (both 147 N, R3); fore-aft 1.5 x 5 kg at 3 g (221 N) on one pin alone; side 1.5 x 5 kg at 1 g with the payload's centre of mass 80 mm below the shoe	R3 and screening values
Materials	6061-T6 yield 276 MPa; 5052-H32 yield 193 MPa; hardened plunger pin 400 MPa in shear (conservative); M4 A2-70 screws at 0.8 x proof (3,161 N)	Handbook values
Swap	Seven hand actions totalling 35 s bare-handed; gloves multiply by 1.5	Estimate
Bus	18 V (6S Li-ion at 3.0 V a cell) to 60 V; 100 W payload; pigtail 0.3 m of 20 AWG, two pins each way; payload switch limit 8 A	KWC-DDR-001, D3
Heat	Lid surface 0.0477 m ² (top and sides); 10 W/m ² K in still air, 25 W/m ² K in hover; 10 K allowance for sun on the frame body around the lid; 10 K from lid air to the hottest part; parts rated -40 to +85 °C	Handbook ranges; maker ratings of the part classes
Dissipation	Flight controller 2.5 W, telemetry 1 W idle and 2 W in flight, receiver 0.3 W, 5.3 V supplies 1.1 W, payload switch 0.47 W, bus conduction 4 W at 100 A through 0.4 mΩ	Estimates; 100 A is Kitewright Lift's hover estimate at 44 V
Atmosphere	International Standard Atmosphere	ISA
Hover model	Hover power proportional to 1/V(density) for the same mass and rotors; unheated Li-ion gives 60 % of capacity at -20 °C; ColdCell packs give 95 % and spend 5 % on their heater	Momentum theory; typical cell data
Mass	Made parts from model volumes (aluminium 2.70, ASA 1.07, G10 1.85, UHMW-PE 0.94 g/cm ³); bought parts from catalogue class values (Table 8)	Estimates, to confirm when bought

■ A. Mount strength (R3)

The vertical design load is 147 N in both R3 cases [A1]. The shoe overlaps each lip by 19 mm along 184 mm; the lip bends about the spacer bar's inner edge with a 10.5 mm lever, giving 2.8 MPa, a factor of 99 on yield [A2]. The ten M4 screws that hold the lips see at most 46 N each with prying, a factor of 69 [A3]. Between the frame bolts, 220 mm apart, the plate's edge strip, the spacer bar and the lip, taken as three separate beams, reach 18 MPa, a factor of 10.8 on the 5052 plate [A4].

Fore and aft, the shoe is held by the front stops one way and by the locking pins both ways. The 221 N fore-aft case on ONE pin gives 11.2 MPa of shear in the 5 mm pin and 8.8 MPa of bearing in the shoe [A5]; forward load into the two front stops gives 1.2 MPa [A6]. A 1 g side load with the payload's centre of mass 80 mm down puts 54 N on each lip [A7]. Every factor is large, because the parts are sized for making and handling, not for strength. That margin is why two independent pins cost little: either pin alone holds the payload.

■ B. Payload swap (R2)

Table 2. Swap sequence [B1].

Action	Time, bare-handed (s)
Unplug the DS-014 plug	6
Pull and twist both locking pins to their rest	5
Slide the payload out to the rear	4
Slide the next payload in to the front stops	5
Twist both pins back; they snap into the shoe	3
Plug the DS-014 plug and close its latch	8
Check both pins show no red band; tug the payload	4
Total	35

With gloves the swap takes about 52 s and needs no tools. The margin to R2's figure is small, so a timed gloved trial at TRL 4 is the real check.

■ C. Payload power (R4)

Table 3. Payload current at the end of discharge [C1].

Bus	Voltage (V)	Current for 100 W (A)	Pigtail drop (mV)
6S Li-ion, 3.0 V a cell	18.0	5.56	56
12S Li-ion	36.0	2.78	28
16S LiFePO4, 2.8 V a cell	44.8	2.23	22

The 8 A payload switch covers the worst case with a 1.44 x margin [C2]. The current each DS-014 battery pin may carry is to be confirmed from the published standard and the chosen connector; two pins each way at 3 A would cover the 6S case.

■ D. Heat (R5)

The lid encloses 4.9 W on the ground and 10.4 W in Kitewright Lift's hover, where 4 W comes from the 100 A bus current [D1]. On a 45 °C day, with 10 K allowed for sun on the frame body, the lid air reaches 65.3 °C on the ground and 63.7 °C in hover, and the hottest part about 75 °C, 10 K inside the 85 °C rating [D2]. Cold-soaked to -20 °C, every part boots within its -40 °C rating, and once running the lid air settles near -10 °C [D3]. The optional companion computer sits outside the lid (KWC-DDR-001, D8), so its 5 W does not count here.

■ E. Altitude (R6)

Table 4. International Standard Atmosphere [E1].

Altitude (m)	Temperature (°C)	Pressure (kPa)	Density (kg/m³)	Share of sea-level density
0	15.0	101.3	1.225	100 %
3,000	-4.5	70.1	0.909	74 %
5,000	-17.5	54.0	0.736	60 %
6,000	-24.0	47.2	0.660	54 %

At 5,000 m and -20 °C the density is 0.743 kg/m³, and the barometer must read down to 54 kPa; flight-controller barometers of this class read 30 to 110 kPa [E2]. The core's share of R6 is sensors in range and failsafes set for each band (Table 6). Whether a frame has the thrust to stay in control is a frame property, checked in Kitewright Lift and Kitewright Range with AltiRig data.

Table 6. Altitude parameter sets: what each band changes (released PX4, parameters only).

Setting	Sea level band (0 to 1,500 m)	3,000 m band (1,500 to 4,000 m)	5,000 m band (4,000 to 6,000 m)
Hover thrust estimate	From the frame's sea-level hover test	Sea-level value x 1.16	Sea-level value x 1.29
Low-battery warning and return	30 % and 20 %	35 % and 25 %	40 % and 30 %
Return altitude above launch	30 m	50 m	80 m (terrain)
Link-loss action	Return after 5 s	Return after 5 s	Return after 3 s
Geofence	Set per site, 1,000 m radius by default	As sea level	As sea level, 120 m ceiling above launch
Pack temperature before arming	ColdCell reports at least 10 °C	As sea level	As sea level

The hover-thrust factors are $1/\sqrt{\text{density share}}$ from Table 4; the battery thresholds are deliberately higher at altitude because a return through thin air takes more energy. All values are starting points to be tuned on AltiRig and in flight.

■ F. Hover-time model (R7)

For the same aircraft and mass, hover power scales with $1/\sqrt{\text{density}}$, so hover time at 5,000 m is 0.775 of sea level from air alone [F1]. Unheated packs that give 60 % of their capacity at -20 °C bring that to 46.5 %, which is the scaffold's 47 % estimate [F2]. ColdCell packs kept warm, giving 95 % of their capacity and spending 5 % on their heater, bring it to 70.0 % [F3]. The model is:

hover time = usable pack energy x (1 - heater share) x efficiency / hover power, with hover power = $(m \cdot g)^{1.5} / (\text{figure of merit} \times \sqrt{2 \times \text{density} \times \text{rotor disc area}})$.

It leaves out the change in motor and propeller efficiency at the higher rotor speeds of thin air; AltiRig measures that. R7 asks that the model predict measured hover time within 15 %, which only TRL 4 flights can show.

■ G. Common core (R8, R10, R1)

Table 7. Outputs needed on the flight controller [G1].

Frame	Outputs needed	Available
Lift, eight motors, plus three payload channels	11	16
Lift, six motors, plus three payload channels	9	16
Range, four lift motors, pusher and five servos	10	16

One core, one frame pattern and one payload interface serve both frames with parameter changes only (R8). Released PX4 (v1.15 or later) runs multirotor and standard VTOL frames, DroneCAN battery reporting and MAVLink payload protocols without source changes (R10). The DS-014 pigtail carries the battery, Ethernet, CAN, UART and trigger lines named in the standard (R1); the exact pinout and licence terms are confirmed from the published standard before the connector is bought.

■ H. Mass (R9)

Table 8. Core mass [H1, H2].

Part	Mass (g)	Source
Core plate	190	Model volume
Rail spacer bars, lips and wear tape	162	Model volume
Front stops, pin blocks, jam nuts, corner spacers	96	Model volume
Lid	114	Model volume
Damping plate	28	Model volume
Flight controller	100	Catalogue class
Power board, power modules	105	Catalogue class
Power leads, 8 AWG, and four AS150 halves	124	Catalogue class
Radio, receiver, antennas and bulkheads	91	Catalogue class
GNSS receiver and mast	45	Catalogue class
Locking pins (2)	50	Catalogue class
DS-014 pigtail, harness, switch, grommets, gasket	110	Catalogue class
Dampers, standoffs, clinch nuts, fasteners	57	Catalogue class
Total (rows rounded)	1,274	

The core weighs about 1.27 kg without the payload shoe (310 g, counted with each payload) and without the optional companion computer (60 g, outside the lid) [H2]. That is 274 g over the figure in R9 [H3]. The structure is heavy because it is sized for making in aluminium sheet and bar, not for strength (section A), and because the four

heavy power leads sit inside the core. Three ways forward, each kept within the concept, are set out for Amish in `docs/REVIEW.md` : a lighter structure (1.12 kg, about USD 60 more) [H4]; the same plus the power leads moved to each frame's harness (0.99 kg) [H5]; or the design as drawn (1.27 kg) [H6].

■ I. Cost (R11)

The bill of materials comes to USD 1,760.60 for the airborne core, USD 1,620.00 for the ground station and USD 560.00 for spares [I1]. Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 3,940.60 (USD 1,059.40 under the target) [I2].

■ L. Results against every requirement

Table 9. Results (also in `docs/04-calcs/results.csv`).

ID	Result	Status
R1	DS-014 40-pin connector on a 0.3 m pigtail; signals from the flight controller and bus	Met on paper (licence and pinout to confirm)
R2	About 52 s with gloves, no tools; pin state shown by a red band	Met on paper (estimate)
R3	Vertical 147 N: lips 99 x, screws 69 x, side beam 10.8 x; fore-aft 221 N on one pin 36 x	Met on paper
R4	100 W needs at most 5.6 A (6S at 18 V); switch limit 8 A	Met on paper (pin rating to confirm)
R5	Hottest part about 75 °C at 45 °C ambient; cold boot at -20 °C within the -40 °C rating	Met on paper
R6	Sensors in range at 54 kPa; failsafes set per altitude band	Met on paper for the core; flight check needs a frame
R7	Model published: 47 % (cold packs) and 70 % (ColdCell) of sea-level hover time at 5,000 m and -20 °C	Model ready; accuracy shown only at TRL 4
R8	One core and one interface; Lift needs 11 of 16 outputs, Range 10	Met by design
R9	1.27 kg estimated	Not met (Decisions for Amish, <code>docs/REVIEW.md</code>)
R10	Released PX4, parameters and stock modules only; ColdCell reports over DroneCAN	Met by design
R11	USD 3,940.60 including ground station and spares	Under the value-engineering target by USD 1,059.40

■ Checks against earlier figures

- The scaffold's estimate of 47 % hover time at 5,000 m and -20 °C is reproduced (46.5 %) and traced to unheated packs; with ColdCell the model gives 70 %.
- R9 was written before the core had a constructable structure; the constructable design does not meet it (section H).